

Adem Toumi

AI Engineer | Data Engineer | Data Analyst

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PROFESSIONAL SUMMARY

AI, Data Science & Data Engineering professional covering the full data spectrum — data analysis, exploratory data analysis (EDA), data wrangling, data pipelines, deep learning, large language models, computer vision, natural language processing, and software development. Graduating July 2026 from ESI-SBA (State Engineer's Degree & M.Sc., Artificial Intelligence & Data Science).

EDUCATION

Higher School of Computer Science (ESI-SBA)

State Engineer's Degree & M.Sc. in Artificial Intelligence and Data Science

Sidi Bel Abbès, Algeria

Sep. 2021 – Jul. 2026

- **Dual Thesis (Engineering & M.Sc.):** *Log-Based Anomaly Detection in Distributed Systems* — research and engineering work submitted for both the State Engineer's Degree and the Master's degree; benchmarked ML, deep learning, and transformer-based approaches across industrial log datasets and built a real-time streaming pipeline with cloud inference and an explainability dashboard.
- **Coursework:** Machine Learning · Deep Learning · NLP · Computer Vision · Distributed Systems · Big Data Engineering · Software Engineering.

TECHNICAL SKILLS

Programming Languages:	Python · SQL · TypeScript · JavaScript · Dart · Java
Data Analysis & EDA:	Data Collection · Data Cleaning · Preprocessing · Data Wrangling · EDA · Visualization · Pandas · NumPy · Matplotlib · Seaborn
Data Engineering:	Apache Kafka · Apache Spark · PySpark · Hadoop · Apache Airflow · Streaming Ingestion Pipelines
Machine Learning:	PyTorch · TensorFlow · Scikit-learn · XGBoost · Supervised & Unsupervised Learning · Model Evaluation
Deep Learning & Vision:	Computer Vision · Object Detection & Tracking · Deep Learning · OpenCV · Edge Hardware
NLP & Generative AI:	Natural Language Processing · Large Language Models · Retrieval-Augmented Generation · Semantic Search · Explainable AI (XAI)
Software & Deployment:	PostgreSQL · Firebase · FastAPI · Node.js · React · Next.js · Flutter · Docker · Git · Linux

RESEARCH & INDUSTRY EXPERIENCE

Research Intern – Computer Vision & Embedded Systems

Feb. 2025 – Jun. 2025

Electronic Systems Research & Development Unit (URD Laboratory), Sidi Bel Abbès, Algeria

- Developed a real-time aerial surveillance system for UAVs — trained object detection models on aerial datasets (mAP@0.5 $\approx$ 0.85), integrated dual tracking pipelines, and deployed the full inference stack on edge hardware (Raspberry Pi 5 + Intel Movidius NCS 2).
- Designed the complete hardware-software architecture: custom UDP video streaming to a Ground Control Station, automated gimbal control that keeps tracked targets centered, and an interactive operator interface with live telemetry.

IT Intern – Database & Network Administration

Sep. 2024

Sonatrach, Direction Maintenance, Biskra, Algeria

- Optimized SQL queries, maintained database schemas, and automated backup scripts in an industrial environment.
- Supported network administration and system monitoring tasks.

SELECTED PROJECTS

Artificial Intelligence, Data Engineering & Analytics

Log-Based Anomaly Detection in Distributed Systems

Python, PyTorch, Kafka, PostgreSQL, Streamlit

- Engineered a complete real-time anomaly detection pipeline for distributed infrastructure: log streams flow through Apache Kafka, ML and transformer-based models run cloud inference in parallel, results persist to a database, and a live dashboard surfaces alerts with per-model metrics.
- Integrated XAI (Explainable AI) using SHAP and LIME — token-level attributions are computed on demand and displayed inline in the dashboard, making every anomaly flag fully interpretable.

KnowledgeHub AI – Enterprise RAG Platform

Python, FastAPI, React, LLMs, Docker

- Built a production-ready document intelligence platform: organizations upload their knowledge base, ask questions in natural language, and receive grounded answers with exact source citations.
- Delivered the complete stack — hybrid search engine, FastAPI backend, React frontend, Docker Compose deployment, and a comprehensive automated test suite.

Object Detection & Tracking for UAV Applications

Python, PyTorch, Computer Vision, Edge AI

- End-to-end aerial surveillance system covering object detection, multi-target tracking, edge deployment, and a Ground Control Station interface — full detail in Research Experience above.

Multi-Class Malicious URL Detection

Python, Scikit-learn, XGBoost, Data Analysis

- Built a multi-class cybersecurity ML classifier extracting character n-grams and structural URL features across 651,191 URLs, delivering 96.8% Macro F1 accuracy.

Paraphrase Detection via Siamese Attention

Python, PyTorch, NLP, Deep Learning

- Designed a sentence-level semantic similarity model using a Siamese Bi-LSTM architecture with multi-head attention trained on PAWS and QQP datasets.

HajjPortal – Trilingual Pilgrimage Management Platform

Next.js 14, TypeScript, Socket.io, Tailwind

- Developed a full-stack, trilingual (Arabic, English, French) web platform with Next.js 14 App Router, dynamic RTL/LTR layout support, JWT authentication, and WebSockets.

KidLink – Cross-Platform School Communication App

Flutter, Dart, Firebase Auth, FCM Push

- Built a cross-platform mobile application in Flutter & Dart with role-based workflows, Firebase FCM push notifications, and Cloud Firestore offline persistence.